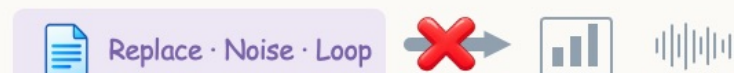


Existing protections address different targets

❌ Solving logic lost. One method cannot cover all three modalities.

Text rewriting



❌ Still not an image or waveform

Private image synthesis



❌ Redrawn, not paired with origin question

Speaker anonymization

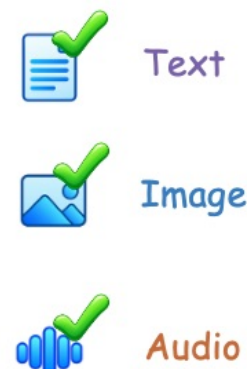
“... formula rate plan increases at Entergy Louisiana ...”

🔄 Same private words, only the voice is new

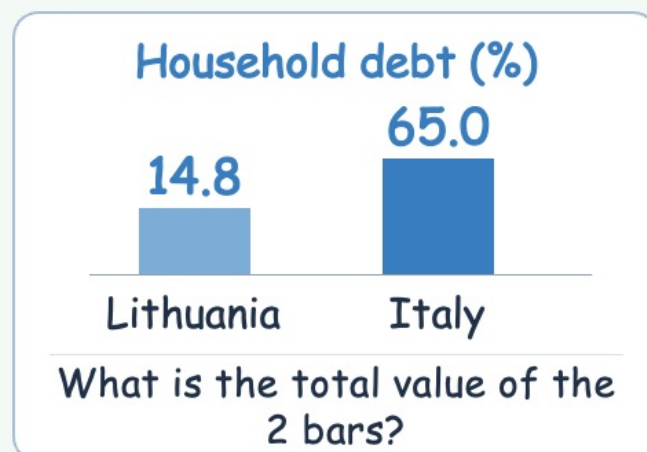
Our Work

✅ Protect private content, keep remote reasoning, one pipeline.

Original inputs

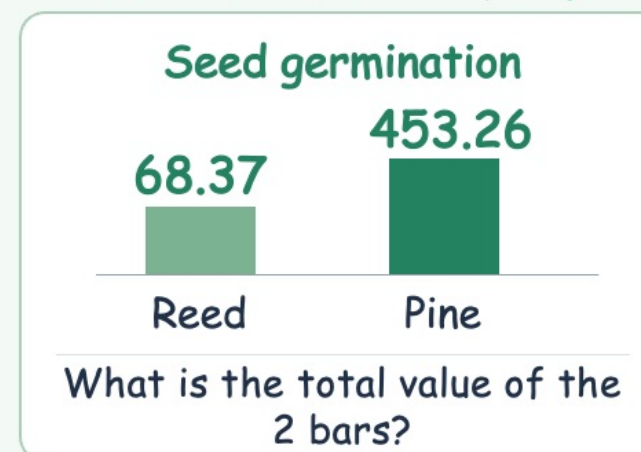


Original stays local



New topic
New names
New values
Same question

Protected media + query



same format

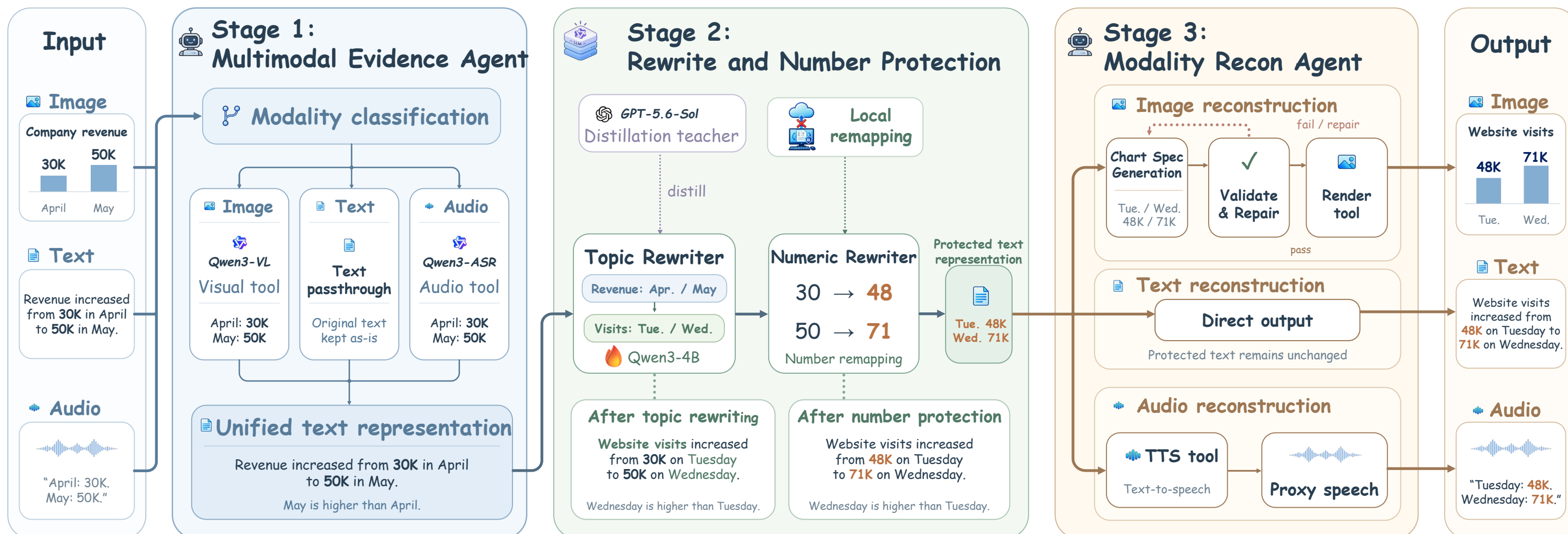
Remote solver

Selects operands and operation
RETURNED PROGRAM
68.37 + 453.26

Execute on original values
14.8 + 65.0 = 79.8

Restore operands locally
68.37 → 14.8 · 453.26 → 65.0

Return the program



Unprotected Query How much did revenue increase from April to May?

Protected query How much did visits increase from Tuesday to Wednesday?